Tutorial: Naive Bayes

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Problem

#	Age	Credit Score	Home Owner	Default?
1	35	High	Yes	No
2	45	Low	No	Yes
3	28	Medium	No	No
4	52	Low	No	Yes
5	38	High	Yes	No
6	41	Medium	Yes	No
7	29	Low	No	Yes
8	55	Medium	No	Yes
9	33	High	Yes	No
10	47	Low	No	Yes

Based on the given data, predict will the following customer default a loan:

Age	Credit Score	Home Owner
40	Low	No

The Naive Bayes Framework

We use **Bayes' Theorem** to find the probability of each class **(C)** given the features **(X)**:

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

The "Naive" Assumption

We assume features are **conditionally independent**. Therefore, the numerator becomes:

$$P(C) \cdot P(\text{Age}|C) \cdot P(\text{Credit}|C) \cdot P(\text{Home}|C)$$

- **Step 1:** Calculate Priors **P(C)**.
- **Step 2:** Calculate Likelihoods for Categorical data (Counting).
- **Step 3:** Calculate Likelihoods for Continuous data (Gaussian PDF).

Step 1 — Prior Probabilities P(C)

- **Total Customers (N):** 10
- **Class "Yes" (Defaulted):** 5

$$P(Default = Yes) = \frac{5}{10} = 0.5$$

- **Class "No" (Did not Default):** 5

$$P(Default = No) = \frac{5}{10} = 0.5$$

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33	High	Yes	No
47	Low	No	Yes

Step 2a — Categorical Likelihoods (Credit Score)

- We analyze the frequency of all categories across the 10 samples.

Likelihoods:

$$P(\text{Low} \mid Yes) = 4/5 = 0.8$$
$$P(\text{Low} \mid No) = 0/5 = 0.0$$

$$P(\text{Medium} \mid No) = 2/5 = 0.4$$
$$P(\text{Medium} \mid Yes) = 1/5 = 0.2$$

$$P(\text{High} \mid No) = 3/5 = 0.6$$
$$P(\text{High} \mid Yes) = 0/5 = 0.0$$

Credit Score	Default = Yes	Default = No
High	0	3
Medium	1	2
Low	4	0

Step 2b — Categorical Likelihoods (Home Owner)

- We analyze the frequency of all categories across the 10 samples.

Likelihoods:

$$P(\text{No} \mid Yes) = 5/5 = 1.0$$
$$P(\text{No} \mid No) = 1/5 = 0.2$$

$$P(\text{Yes} \mid Yes) = 0/5 = 0$$
$$P(Yes \mid No) = 4/5 = 0.8$$

Home Owner	Default = Yes	Default = No
Yes	0	4
No	5	1

Step 3 — Continuous Likelihood (Age)

Assuming Gaussian Distribution:

$$P(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- **1. Calculate Statistics for Age:**
  - **Yes:** $\mu = 45.6,\ \sigma^2 = 81.44$
  - **No:** $\mu = 35.0,\ \sigma^2 = 19.6$

Predicting for the New Customer

Age	Credit Score	Home Owner
40	Low	No

The Calculation Formula (Naive Bayes):

$$P(Class \mid Features) \propto P(Class) \times P(\text{Age} \mid Class) \times P(\text{Credit} \mid Class) \times P(\text{Home} \mid Class)$$

Case 1 — Probability of "Default = Yes"

- **Prior P(Yes):** 0.5
- **Likelihood P(Age=40 | Yes):** 0.036
- **Likelihood P(Credit=Low | Yes):** 0.8
- **Likelihood P(Home=No | Yes):** 1.0

P(Default = Yes | new customer):

$$0.5 \times 0.036 \times 0.8 \times 1.0 = \mathbf{0.0144}$$

Case 2 — Probability of "Default = No"

- **Prior P(No):** 0.5
- **Likelihood P(Age=40 | No):** 0.047
- **Likelihood P(Credit=Low | No):** 0.0
- **Likelihood P(Home=No | No):** 0.2

P(Default = No | new customer):

$$0.5 \times 0.047 \times 0.0 \times 0.2 = \mathbf{0.0}$$

The "Zero-Frequency" Problem

The Issue:

In our manual calculation for **Default = No**, we encountered:

$$P(\text{Credit=Low} \mid No) = \mathbf{0}$$

Because Naive Bayes multiplies all likelihoods, a single zero determines the outcome.

- It doesn't matter if the **Age** (40) was a perfect match for the "No" class.
- The result becomes:

$$0.5 \times 0.047 \times \mathbf{0} \times 0.2 = \mathbf{0}$$

The Solution — Laplace Smoothing

To fix this, we add two "Dummy Rows" to our dataset:

- One **Default = Yes** who has every feature value once.
- One **Default = No** who has every feature value once.

Revised Likelihood Formula ($+1$ Smoothing):

$$P(x_i \mid c) = \frac{\text{Actual Count} + 1}{\text{Total Class Samples} + \text{Number of Categories}}$$

The Solution — Laplace Smoothing

Recalculated Likelihoods for our Target:

Credit = Low:

$$P(Low \mid Yes) = \frac{4+1}{5+3} = \frac{5}{8} = \mathbf{0.625}$$

$$P(Low \mid No) = \frac{0+1}{5+3} = \frac{1}{8} = \mathbf{0.125}$$

Home = No:

$$P(No \mid No) = \frac{1+1}{5+2} = \frac{2}{7} \approx \mathbf{0.286}$$

$$P(No \mid Yes) = \frac{5+1}{5+2} = \frac{6}{7} \approx \mathbf{0.857}$$

Revised Score (Yes): $0.5 \times 0.036 \times 0.625 \times 0.857 \approx \mathbf{0.0096}$

Revised Score (No): $0.5 \times 0.047 \times 0.125 \times 0.286 \approx \mathbf{0.0008}$

The prediction

Default = Yes